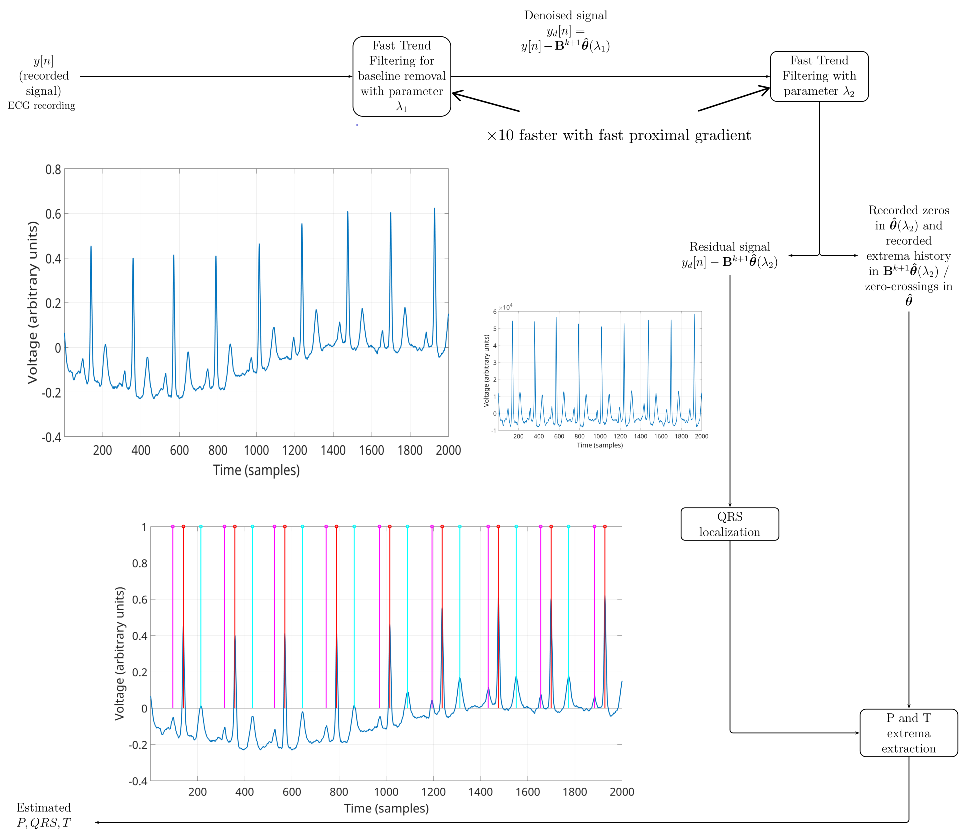
Graphical Abstract

FDP-TF: A Fast Two-Pass Trend Filtering for ECG Delineation

Tom Trigano, Yann Sepulcre, Matan Masika, Asher Perez, David Luengo



Fast Double Pass Trend Filtering (FDP-TF) pipeline

Highlights

FDP-TF: A Fast Two-Pass Trend Filtering for ECG Delineation

Tom Trigano, Yann Sepulcre, Matan Masika, Asher Perez, David Luengo

- We present an improvement of the Adaptive Trend Filtering for ECG delineation.
- The new method is over 10× faster than Adaptive Trend Filtering.
- The method can be efficiently used for long signals or real-time applications.
- Results obtained on peak localization are on par with state-of-the-art method.

FDP-TF: A Fast Two-Pass Trend Filtering for ECG Delineation

Tom Trigano[a,*], Yann Sepulcre[b], Matan Masika[a], Asher Perez[c], David Luengo[d]

[a]*Shamoon College of Engineering, Department of Electrical Engineering, Jabotinsky St, 84, Ashdod Campus, Ashdod, Israel*
[b]*Sapir Academic College, Department of Mathematics, Hof Ashkelon, Sderot, Israel*
[c]*Jerusalem College of Technology, Department of Electrical and Electronic Engineering, Jerusalem, Israel*
[d]*Polytechnic University of Madrid, Department of Audiovisual and Communications Engineering, Madrid, Spain*

Abstract

ECG delineation, which involves detecting key waveform features such as P, QRS, and T waves, is essential for accurate cardiac monitoring and diagnosis. In a recent study, the authors introduced the Adaptive Trend Filtering (ATF) algorithm, which effectively identifies local extrema in ECG signals for both denoising and delineation, demonstrating strong performance compared to state-of-the-art methods. However, its implementation using the Alternating Direction Method of Multipliers (ADMM) resulted in long execution times, limiting its practical application. This paper presents Fast Double-Pass Trend Filtering (FDP-TF), a complete reformulation of the original ATF algorithm that leads to a significantly faster and novel algorithm. The results show that the novel FDP-TF algorithm attains a speedup of several orders of magnitude with no loss in algorithm performance, thus making it a compelling alternative for fast off-line processing of very long ECG recordings, such as those encountered in the QT dataset, real-time processing of on-line signals, and even implementation in wearable devices. The results also show that this increase in speed is achieved without compromising the performance of the FDP-TF algorithm, which remains on par with current state-of-the-art leading approaches.

*Corresponding author
Email address: thomast@ac.sce.ac.il (Tom Trigano)

Keywords:
ECG signal processing, compressive sensing, adaptive trend filtering, delineation, QT dataset.

1. Introduction

The analysis of electrocardiographic (ECG) signals, which capture the heart's electrical activity, has been extensively studied, as understanding their structure and relationship to the underlying electrical phenomena inside the heart is crucial for accurate diagnosis [1, 2, 3, 4].

In this context, ECG delineation plays a critical role; its aim is to provide precise algorithms working in real-time which can segment ECG recordings and locate all the relevant peaks of the P, QRS and T waves, in order to facilitate the diagnosis. Indeed, ECG denoising strategies and efficient peak localization methods can improve further classification performance for diagnosis tasks [5]. This specific task is hindered by inter/intra-subject variability (i.e., changes in morphology, duration and relative lags among the relevant waves in the ECG signals), as well as the presence of different types of noises and interferences. Regarding noise, four common disturbances can be identified in external ECG recordings: baseline wander (low-frequency activity from sources like body movement and respiration), electrode motion (noise caused by changes in electrode-skin impedance due to movement), muscle artifacts (i.e., electromyographic (EMG) interference from muscular electrical activity), and powerline interference from nearby electrical equipment.

Due to the previous considerations, ECG delineation remains a very active field of research, even though it has been studied for a long time.

1.1. Previous works

ECG delineation has been continuously investigated, particularly since the development of digital signal processing (DSP) techniques. For example, [6] combines denoising by means of bandpass filtering and derivative-based QRS detection and uses zero-crossing to detect the other wave extrema. The digital filters involved in the process are usually user-defined [7]; however, a more modern approach described in [8] suggests to learn the finite impulse response transform of the linear filters involved in delineation by considering them as 1D convolutional layers of a deep neural network.

DSP-based methods have long served as the foundation for ECG delineation due to their computational efficiency, interpretability, and ability to incorporate domain-specific knowledge such as characteristic frequency bands and temporal relationships. However, these approaches exhibit some limitations: they often rely on fixed parameters that are sensitive to inter- and intra-subject variability, and they struggle in the presence of nonstationary noise, overlapping waves, or atypical morphologies caused by arrhythmias or pathologies. While DSP remains an important component in many delineation pipelines, especially for its low latency and physiological grounding, it is increasingly being augmented or supplanted by data-driven or hybrid approaches to improve accuracy. Other approaches distinct from linear filtering also provide excellent performance. For example, [9] shows that the phasor transform can lead to a fast and efficient delineation of the QRS, P and T waves of the ECG. In [10], after appropriately filtering the ECG, the authors exploit an optimal area property of some mobile trapezium lying on the waveform after the (already estimated) T-peak position in order to detect specifically the T-wave end.

More recent signal processing approaches rely on wavelets, which allow both to remove some types of noise and to capture typical ECG morphologies at different levels of analysis. The use of the Continuous Wavelet Transform (CWT) to capture fiducial points sequentially (first the QRS based on the maxima of the CWT modulus, and then the T and P waves) and their onsets is shown in [11] and [12]. However, CWT methods can be computationally intensive, especially when high time-frequency resolution is required, making real-time implementation challenging without significant optimization. Therefore, the use of the Discrete Wavelet Transform (DWT) is often preferred for real-time applications. As shown in numerous works (e.g., [13]), the analysis of the levels of the DWT provides insights on the fiducial points with very good precision. Specific properties of some multiscale discrete wavelet decompositions related to the ECG wave structure (such as zero-crossings and extrema locations of some coefficients) are also used to successfully extract fiducial points, e.g. in [14]. In [15], an undecimated multiscale wavelet decomposition is applied to different segments of the ECG as a feature extraction step, in order to train hidden Markov and hidden semi-Markov models (the last one incorporating a more appropriate state duration modelling). In [16], based on an interpretation of fiducial points as local minima or maxima of morphological derivative difference operators, the authors use a multiscale morphological derivative transform, similar in its concept to a multiresolution wavelet decomposition, of the ECG signal with good performances on QRS complex as well as on onsets and offsets of the characteristic waves. DWT-based methods are efficient and well-suited for real-time ECG delineation due to their multiscale decomposition and low computational cost. However, they have a more limited temporal resolution than their CWT counterparts (thus making the P and T waves analysis more challenging), due do their fixed dyadic scale structure.

The development of compressive sensing and associated methods allowed to look at ECG delineation as a sparse reconstruction problem, where enforced sparsity can be used to extract features of interest from ECG recordings. For intra-cardiac ECG recordings, [17] suggested a post-processed version of the LASSO regressor with a dictionary made of Mexican hat wavelets for peak localization, and an overlapping Group LASSO with a dictionary of Gaussian shapes was proposed in [18]. Using the same paradigm, [19, 20] suggested replacing the ℓ_1-norm by another penalty, taking into account the refractory period to achieve better results. Indeed, sparse regression based techniques offer a model-based approach to ECG delineation, enabling the incorporation of prior knowledge and smoothness constraints and yielding directly interpretable results. However, their effectiveness depends heavily on the choice of the basis functions and regularization parameters, and often require preliminary processing. Furthermore, their use for very long recordings needs extensive memory use and is computationally intensive. Additional methods relying on signal processing or classical machine learning are also described in the review paper [21].

Data-driven learning strategies were also investigated for delineation. For example, a bi-directional hidden semi-Markov model was trained on preprocessed features such as the homomorphic envelope or the Power Spectral Density envelope to infer the locations of the fiducial point in [22]. However, most supervised learning approaches rely nowadays on Deep Neural Networks (DNNs), as they do not require complex feature engineering. In most publications using this paradigm, ECG delineation is considered as a blind segmentation problem, where the objective is to return peak areas instead of fiducial points. In [23] and [24], the authors rely on encoder-decoder architectures to extract sequentially the ECG pulses areas and to split the obtained areas into peaks of interest. On the other hand, both [25] and [26] consider ECG recordings as time series, and suggest convolutional layers for feature extraction, followed by an LSTM network to extract the peaks onsets and offsets sequentially. DNNs can also be used for fiducial points extraction. For instance, [27] uses a convolutional neural network and an LSTM network for feature extraction, which are afterwards used to train an ensemble learner to detect fiducial points of the QRS complex and T waves. Similarly, [28] uses a ResNet architecture followed by an LSTM layer to extract waves peaks with excellent precision. In more recent contributions involving deep learning, [29] and [30] considered ECG delineation as a segmentation problem; this segmentation was addressed using an encoder-decoder architecture in [29], and [31] compared four architectures based on encoder-decoder convolutional neural networks, investigating whether the addition of recurrent layers and residual connections between feature maps could improve performance. A two-stage approach involving feature extraction using stacked convolutional layers and a recurrent BiLSTM architecture was also proposed for single lead ECG delineation in [32], and was extended to 12-lead recordings in [33]. More recently, recurrent layers have been replaced by transformers (e.g., in [34]) with excellent delineation results, and variational encoders have been suggested to take into account the inherent morphological variability of ECG recordings [35]. DNN based approaches have shown strong performance in ECG delineation, often outperforming traditional methods. However, DNNs require large, well-annotated datasets to generalize reliably, and their black-box nature poses challenges for clinical interpretability and regulatory acceptance. They are also computationally intensive, making real-time deployment and energy-efficient implementation difficult, particularly in embedded or wearable systems.

1.2. Proposed approach

In a recent publication [36], we proposed the Adaptive Trend Filtering (ATF) as an alternative for ECG denoising and peaks localization. This approach includes two stages: the first step removes the baseline using the Trend Filtering (see [37]), and is solved numerically by means of the Alternating Direction Method of Multipliers (ADMM), described, e.g., in [38]; the second step iterates the ADMM with varying regularization parameters, saves the local extrema at each iteration, and estimates the position of the P, QRS and T waves extrema (we refer to [36] for a detailed description of ATF). Though showing results comparable to state-of-the-art methods, the main shortcoming of the ATF is its speed of execution. Indeed, the ADMM implementation for Trend Filtering requires numerous matrix-vector multiplications, and storing large matrices in memory to obtain results in a reasonable time. In particular, this last point prevents the use of ATF on long ECG recordings: even though the signal could be split into smaller parts before processing, the resulting execution time would be too long for a real-life application. Numerous methods (e.g., see [39]) have been suggested to accelerate the convergence of Trend Filtering; nevertheless, the proposed methods focus more on the convergence acceleration rather than the execution time, and therefore still include numerous matrix operations, which are extremely time-consuming for large recordings. In this paper, we present a Fast Double-Pass Trend Filtering (FDP-TF) method, involving a complete reformulation of the original ATF algorithm that leads to a significantly faster and novel algorithm. The proposed FDP-TF is conceived as a fast algorithm, relevent for real-time processing of long recordings, without any training involved and satisfactory precision. In constrast to DSP, regression and wavelet based techniques, it does not rely on any user-defined basic shapes, but rather on the signals' inner regularity properties. It significantly improves ATF in terms of speed, and compares well with other state-of-the-art approaches, as seen in the summary Table 1 and the obtained results.

Method (reference)	User-defined parameters	Requires Training	Data-driven	Suited for real-time	Suitable for long signals	Pre/Post-processing involved
[6]	Many	No	No	Yes	Yes	Yes
[31]	Few	Yes	No	Yes	Yes	No
[11]	Many	No	No	No	No	No
[12]	Many	No	No	No	No	No
[13]	Many	No	No	Yes	Yes	No
[17],[18]	Many	No	No	No	No	Yes
[20]	Many	No	No	No	No	Yes
[22]	Few	Yes	Yes	No	Yes	Yes
[23] to [35]	Few	Yes	Yes	Yes	Yes	Yes
[36]	Few	No	Yes	No	No	Yes
Ours	**Few**	**No**	**Yes**	**Yes**	**Yes**	**Yes**

Table 1: Summary of the advantages and shortcomings of existing methods

The main contributions of this paper are detailed below:

1. We show that a reformulation of the ordinary Trend Filtering allows us to obtain an algorithm which does not require storing any matrix in memory besides the ECG recording. In contrast to the previous version of ATF, where large matrices had to be stored in memory and loaded at each iteration, this reformulation allows the processing of very long recordings.

2. This reformulation also allows us to solve the optimization problems involved in ATF by means of a fast proximal gradient approach [40], instead of ADMM. In this paper, we provide additional insight to optimally set the gradient step and to perform the proximal gradient iterations efficiently, resulting in a significant gain in speed. The localization of the peak extrema also relies on a faster criterion allowed by the previous reformulation.
3. We demonstrate that FDP-TF outperforms ATF in terms of execution time by one order of magnitude, using ECG recordings simulated using the ECGSYN models. Furthermore, this is achieved without compromising the resulting accuracy, as shown in the results obtained on the QT dataset from PhysioBank [41]. We also show that the GPU implementation of FDP-TF allows us to attain results in quasi real-time, thus paving the way for the implementation of the proposed approach in wearable devices in the future.

In summary, FDP-TF addresses the problem of accurately finding the peaks of the relevant waves in the ECG in real-time without the large computational cost and memory storage required by deep learning based methods.

The rest of this paper is organized as follows. Section II reviews the ATF algorithm. Section III details our reformulation and describes the FDP-TF approach in depth. Section IV reports simulation results, compares FDP-TF to state-of-the-art methods in peak detection, and provides an execution time analysis. Section V concludes the paper, discussing potential further development.

2. Overview of Adaptive Trend Filtering

2.1. Model Description

Throughout this paper, we investigate the recorded ECG signals $y[n]$ using the following model:

$$y[n] = s[n] + r[n] + \varepsilon[n], \; n = 1 \ldots N.$$

In Equation (**??**), $s[n]$ denotes a slowly time-varying signal representing the baseline of the ECG recording and the respiration noise, $r[n]$ denotes a more-rapidly varying signal representing the ECG pulses that we wish to retrieve, and $\varepsilon[n]$ is an additional noise term including all the perturbations with higher frequencies (e.g., EMG noise, electrode motion artifacts, and Gaussian noise). Obviously, this model can be vectorized as

$$\mathbf{y} = \mathbf{s} + \mathbf{r} + \boldsymbol{\varepsilon}$$

where the vectors $\mathbf{y}$, $\mathbf{s}$, $\mathbf{r}$, and $\boldsymbol{\varepsilon}$ are N-dimensional, denoting respectively the overall recorded signal, and the separate contributions of the baseline wander, the ECG pulses, and the higher-frequency additive noise. Using (**??**), the general objective of any delineation algorithm is to extract $\mathbf{r}$ from $\mathbf{y}$, while introducing as little distortion as possible in $\mathbf{r}$, and to locate relevant features from the obtained estimate of $\mathbf{r}$. In this paper, we will focus on the peak finding task: locating the local maxima of the P, QRS, and T waves in the obtained noisy recordings.

If the waveforms of the ECG signals were perfectly known, a possible approach to address the delineation problem would be to build a dictionary from these known shapes, $\mathbf{A}$, and build a sparse representation of $\mathbf{y}$ in which the non-zero entries of the regressor would indicate each waveform's location. Such methods have been investigated in previous contributions [17, 19]. However, these approaches strongly rely on the fact that the dictionary is precisely defined, which can be challenging in most biomedical applications. In the following we describe the proposed alternative, which is based on Trend Filtering, and does not require the definition of any dictionary.

2.2. Adaptive Trend Filtering Overview

Adaptive Trend Filtering (ATF) was proposed by the authors in [36] as an unsupervised method for denoising and automatic localization of the peaks of the P, QRS and T waves. A graphical summary of ATF can be found in Figure 1.

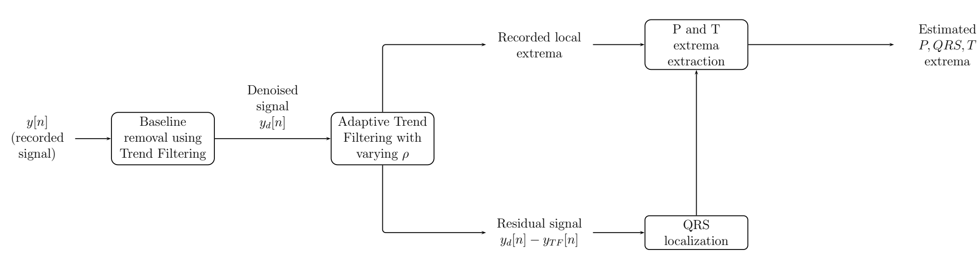


Figure 1: Graphical description of the ATF algorithm

ATF includes two steps. The first step removes the baseline from the ECG recording using a regular trend filtering (e.g., see [42] and [37]), returning a denoised signal $y_d[n]$. The second step consists in an adaptive version of trend filtering (performed on $y_d[n]$), which is solved using ADMM with a slowly decaying regularization parameter ρ_n, and returns a smooth reconstruction $y_{TF}[n]$. At each iteration, local extrema are saved. The QRS complexes are located based on the residual error $y_d[n] - y_{TF}[n]$, and are used altogether with the local extrema history to find the maxima of the P and T waves. Since trend filtering is central to both steps, we briefly define it below.

Given a user-specified order k ≥ 0 and sparsity parameter λ, the trend filtering estimate is defined as

$$\hat{\mathbf{x}}^{\mathbf{trend}}(\lambda) = \arg\min_{\mathbf{x}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{D}^{k+1}\mathbf{x}\|_1 \right\},$$

where $\mathbf{D}^{k+1} \in \mathbf{R}^{N\times N}$ is the discrete difference operator of order $k+1$, which can be recursively defined as

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & & \ddots & \ddots & \\ 0 & 0 & \dots & -1 & 1 \end{bmatrix}$$

and $\mathbf{D}^{k+1} = \mathbf{D}\mathbf{D}^k$ for $k \geq 1$. As shown in [37], the penalty term $\|\mathbf{D}^{\mathbf{k}}\mathbf{x}\|_1$ introduced in (**??**) enforces the solution $\hat{\mathbf{x}}^{\mathbf{trend}}(\lambda)$ to exhibit a piecewise polynomial behavior, where the number of knots between the different pieces decreases as λ increases. Thus, trend filtering can be seen as a sparse penalty applied on the functional smoothness of the result. This property makes it suitable to smooth noisy data, and to estimate the localization of portions of signals with different rates of variation.

2.3. Numerical Implementation of ATF Using ADMM

The trend filtering steps in ATF are numerically computed via ADMM. Given the trend filtering formulation in (**??**), the optimization problem becomes

$$\begin{aligned} &\text{minimize} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{z}\|_1 \right\} \\ &\text{subject to } \mathbf{D}^{k+1}\mathbf{x} - \mathbf{z} = \mathbf{0}, \end{aligned}$$

and the associated ADMM iterations are (see e.g. [38]):

$$\begin{aligned}
\mathbf{x}^{(n+1)} &= (\mathbf{I} + \rho_n (\mathbf{D}^{k+1})^T \mathbf{D}^{k+1})^{-1} \\
&\quad \times (\mathbf{y} + \rho_n (\mathbf{D}^{k+1})^T (\mathbf{z}^{(n)} - \mathbf{u}^{(n)})) \\
\mathbf{z}^{(n+1)} &= \mathbf{S}_{\lambda/\rho_n}(\mathbf{D}^{k+1}\mathbf{x}^{(n+1)} + \mathbf{u}^{(n)}) \\
\mathbf{u}^{(n+1)} &= \mathbf{u}^{(n)} + \mathbf{D}^{k+1}\mathbf{x}^{(n+1)} - \mathbf{z}^{(n+1)}
\end{aligned}$$

where $\mathbf{S}_a$ in (**??**) is the soft thresholding operator, defined by

$$\mathbf{S}_a(\mathbf{x}) \triangleq (|\mathbf{x}| - \alpha)_+ \mathrm{sign}(\mathbf{x}) = \Big[\max\{|x_i| - \alpha, 0\}\mathrm{sign}(x_i)\Big]_{1 \le i \le N}$$

Although straightforward, this formulation involves numerous time-consuming operations. More specifically:

- Steps (**??**) and (**??**) include a matrix-vector multiplication. Though less time-consuming than ADMM's first step, storing the matrix $\mathbf{D}^{k+1}$ in memory can be very challenging for long ECG recordings such as the ones present in the QT dataset.
- More problematically, step (**??**) of ADMM involves matrix-matrix multiplications and a matrix inversion. Though the inverse matrices $(\mathbf{I} + \rho_n(\mathbf{D}^{k+1})^T\mathbf{D}^{k+1})^{-1}$ can be precomputed and stored, these computations and loadings at each iteration are extremely time-consuming, and require a large amount of memory. Furthermore, since ATF employs a slowly decaying regularization parameter ρ_n at each iteration, the number of stored matrices is equal to the number of iterations, involving considerable computational and memory requirements.

These considerations motivate the need for faster or more memory-efficient methods when dealing with lengthy ECG signals.

3. Fast Double-Pass Trend Filtering (FDP-TF): Accelerated Version of ATF for ECG Peak Detection

As previously described, the practical implementation of ATF includes a computational bottleneck, which prevents its use for real-life applications. Since the final steps in ATF have linear complexity, it is therefore essential to make the steps relying on Trend Filtering more efficient. In this section, we introduce the Fast Double-Pass Trend Filtering (FDP-TF) algorithm, specifically designed to address this issue.

3.1. Trend Filtering Reparametrization and Acceleration

Following [43], we note that after a reformulation, solving (**??**) is equivalent to solving the following problem:

$$\begin{cases} \widehat{\mathbf{x}}^{\mathbf{trend}}(\lambda) & = \mathbf{B}^{k+1}\hat{\boldsymbol{\theta}}(\lambda), \\ \hat{\boldsymbol{\theta}}(\lambda) & = \arg\min_{\boldsymbol{\theta}} \left\{ \frac{1}{2}\|\mathbf{y} - \mathbf{B}^{k+1}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_1, \right\}, \end{cases}$$

where $\mathbf{B}$ is the lower triangular matrix whose lower entries are equal to 1. Indeed, (**??**) can be solved in place of the regular trend filtering formulation, since $\mathbf{B}^{-1} = \mathbf{D}$. In this form, the problem becomes the minimization of the well-known Least Absolute Shrinkage and Selection Operator (LASSO) introduced in [44].

The reformulation introduced in (**??**) offers numerous advantages. First, it can be solved more efficiently using proximal gradient methods [40], since the proximal operator has a closed form in this case; it also means that much faster rates of convergence can be attained, for example using the FISTA approach [45]. More importantly, we note that for any vector

$$\mathbf{x} = \begin{bmatrix} x_1, & x_2, & \cdots & x_N \end{bmatrix}^\top;$$
$$\mathbf{B}\mathbf{x} = \begin{bmatrix} x_1, & x_1 + x_2, & \cdots, & \displaystyle\sum_{k=1}^{n} x_k \end{bmatrix}^\top$$

and

$$\mathbf{B}^\top\mathbf{x} = \left[\sum_{k=1}^{n} x_k, \sum_{k=2}^{n} x_k, \cdots, x_n\right]^\top.$$

Therefore, in practice all the matrix-vector multiplications involved in the gradient computation can be done using accumulations, without any matrix storage and computations involved.

Solving the optimization problem described in (**??**) is done by iteratively applying the proximal operator on an initial $\boldsymbol{\theta}$ until convergence [45]). This process relies on iterations of a soft-thresholding gradient. In the case of LASSO, we compute until convergence, at each iteration m,

$$\boldsymbol{\theta}^{(m+1)} = \mathbf{S}_{\lambda\mu_m}\left(\boldsymbol{\theta}^{(m)} + \mu_m(\mathbf{B}^{k+1})^T(\mathbf{y} - \mathbf{B}^{k+1}\boldsymbol{\theta}^{(m)})\right),$$

where $\mathbf{S}_\alpha$ is the soft-thresholding operator introduced in (**??**), λ is the sparsity parameter used in the LASSO, and μ_m is a gradient step. As mentioned before, the matrix-vector multiplication in (**??**) can be performed in that specific case with a linear complexity $O(N)$ by the following simple observation: the vector $\mathbf{B}^{k+1}\boldsymbol{\theta}^{(m)}$ is computed in $(k+1)\,N$ time (since left multiplying by $\mathbf{B}$ costs N computations); left multiplying $\mathbf{y}-\mathbf{B}^{k+1}\boldsymbol{\theta}^{(m)}$ by $(\mathbf{B}^{k+1})^T$ costs an additional $(k+1)\,N$ time, so the term $(\mathbf{B}^{k+1})^T(\mathbf{y}-\mathbf{B}^{k+1}\boldsymbol{\theta}^{(m)})$ in (**??**) costs an overall $2(k+1)\,N$ computation steps, and hence is $O(N)$.

The gradient step μ_m must be carefully chosen to ensure fast convergence. It is well known (e.g., see [46]) that the maximal step allowed in the proximal gradient method is

$$\mu_m = \mathrm{Sp}\Big((\mathbf{B}^{k+1})^\top \mathbf{B}^{k+1}\Big)^{-1},$$

where $\mathrm{Sp}(\cdot)$ denotes the largest eigenvalue of a matrix. As seen in [45], μ_m is the inverse of the Lipschitz constant of the gradient of $\frac{1}{2}\|\mathbf{y}-\mathbf{B}^{k+1}\boldsymbol{\theta}\|_2^2$. Since $(\mathbf{B}^{\mathbf{k+1}})^\top\mathbf{B}^{k+1}$ is large, in practice we cannot use any decomposition technique to compute the gradient step. However, it can be approximated using power iterations. More precisely, if $\mathbf{b}_0$ denotes a random, non-zero vector with the same size as $\boldsymbol{\theta}$, a good approximation of an eigenvector associated with the largest eigenvalue of $(\mathbf{B}^{k+1})^\top\mathbf{B}^{k+1}$ can be computed by

$$\mathbf{b}_{n+1} = \frac{(\mathbf{B}^{k+1})^\top\mathbf{B}^{k+1}\mathbf{b}_n}{\|(\mathbf{B}^{k+1})^\top\mathbf{B}^{k+1}\mathbf{b}_n\|_2}.$$

Since $\mathbf{B}^T\mathbf{B}$ is a Gram matrix, convergence of the power iterations in (**??**) to its largest eigenvector is guaranteed. Furthermore, the computation in (**??**) can be performed following the same approach as before.

3.2. Description of Fast Double-Pass Trend Filtering for ECG Peak Localization

In light of the previously described reformulation, we suggest to update ATF to FDP-TF as described in Figure 2. Besides its acceleration, the first step (applying trend filtering to remove the baseline) remains unchanged. Obviously, the original AdapTrend could be used on $\mathbf{B}^{k+1}\hat{\boldsymbol{\theta}}(\lambda_2)$. However, we suggest here an alternative and more efficient approach. In contrast to ATF, in FDP-TF we can also use the fact that $\boldsymbol{\theta}$ alone can be used to localize the peaks. Indeed, if we set $k=0$ in (**??**), the LASSO parametrization provides a sparse representation of the derivative; this allows us to locate local extrema

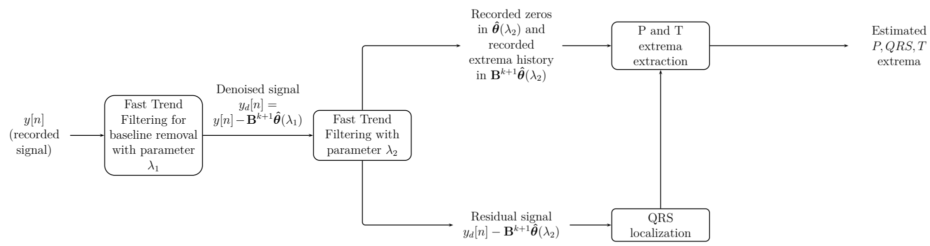


Figure 2: Graphical description of FDP-TF

by finding the sign changes in the pattern of $\hat{\boldsymbol{\theta}}(\lambda_2)$ (by sign change at point i we mean that $\hat{\theta}_{i-1}\hat{\theta}_{i+1} < 0$ does hold for $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\lambda_2)$).

Figure 3 shows a part of a signal taken from of the QT dataset [47], with the obtained $\hat{\boldsymbol{\theta}}(\lambda_2)$ for 30,000 iterations of the aforementioned accelerated trend filtering. A comparison between the signal and the numerical zeros of $\hat{\boldsymbol{\theta}}(\lambda_2)$ (red dotted points shown directly on the signal for a better comparison) illustrates that the peak localization could be performed directly from the LASSO solution.

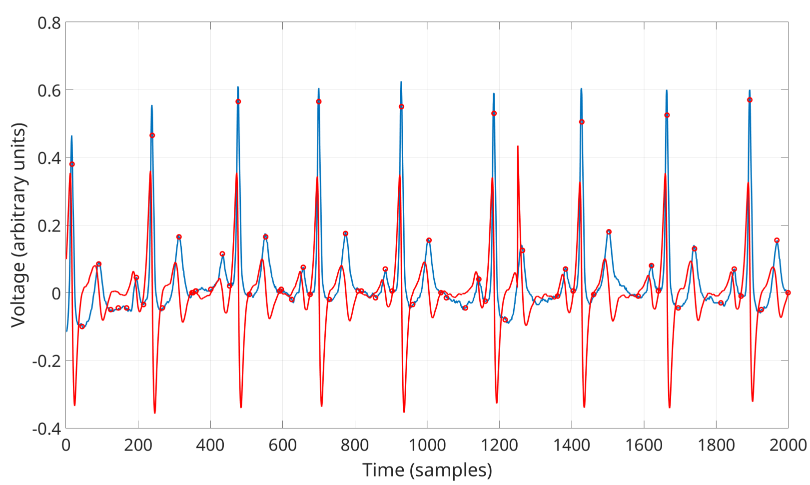


Figure 3: Signal from the QT dataset (blue curve) and obtained $\hat{\boldsymbol{\theta}}(\lambda_2)$ after 30000 iterations of the proposed approach (red curve). The zeros' locations of $\hat{\boldsymbol{\theta}}(\lambda_2)$ are reported with red dots on the signal.

Similarly to the AdapTrend approach, the number of iterations in the second pass plays an important role. Indeed, at early stages, only the most significant sign changes between coefficients of $\hat{\boldsymbol{\theta}}(\lambda_2)$ (occurring at the QRS complexes) are detected, since LASSO tends to provide a sparse solution. However, as the iterations of (**??**) continue, further sign changes are detected at the P and T waves peaks locations, while for a large number of iterations the sign changes related to the noise may also be detected. This behavior is further illustrated in Figure 4, which displays the position of the sign changes as a function of the number of iterations for 2000 sampling points of a signal from the QT dataset. As Figure 4 clearly illustrates, the history of sign changes in $\hat{\boldsymbol{\theta}}(\lambda_2)$ can be used instead of the history of local maxima used in [36].

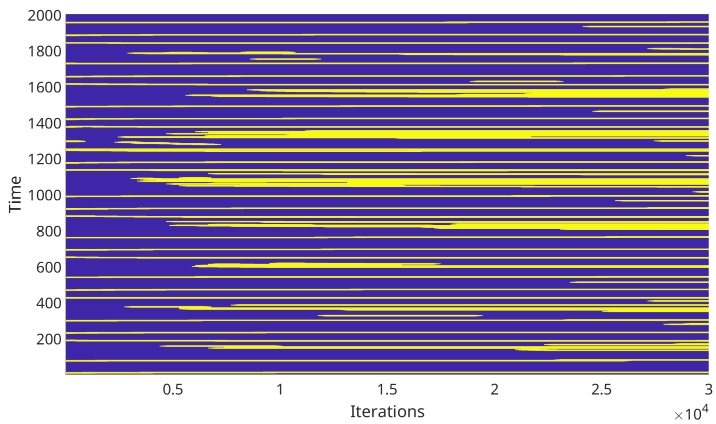


Figure 4: Map of the detected zeros - The biggest changes in the derivative are related to the QRS, and therefore are detected at early stages. T wave and P wave peaks are detected at a later stage

Furthermore, when comparing the averages of the active zero-crossings over the iterations with the signal as in Figure 5, we can observe that the highest values of these averages are well aligned with the peak locations. These considerations show that the peaks localization can indeed rely on the sign changes of $\hat{\boldsymbol{\theta}}(\lambda_2)$ rather than the extrema of $\mathbf{B}^{k+1}\hat{\boldsymbol{\theta}}(\lambda_2)$.

4. Results and Discussion

This section presents the results obtained using FDP-TF for peak localization. We investigated FDP-TF from two different points of view: the gain of speed brought by the proposed acceleration on the Trend Filtering part,

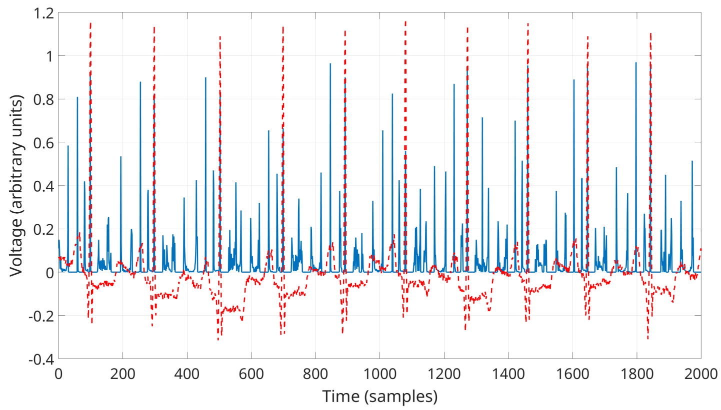


Figure 5: Part of a signal from the QT dataset (red dotted curve) and average of the zero locations recorded at each iteration of the proposed method (overall 20000).

and the peak localization precision on the QT dataset [47]. This dataset includes ECG data for 105 patients recorded at a sampling rate of 250 Hz over a duration of 15 minutes for each recording.

4.1. Simulations Results

Since we showed that the main bottleneck in the AdapTrend method is the implementation of the Trend Filtering using ADMM, we focused on this specific part in the presented simulations. We investigated the computation times of both versions (regular and accelerated) for signals of varying sizes, ranging from $N = 100$ samples to $N = 10^6$ samples. For all sizes considered, we generated 100 synthetic ECG signals based on the well-known ECGSYN model [48]. We apply on each signal 10000 iterations of both versions of the Trend Filtering, and recorded both execution times. The experiments were carried out on a desktop computer with an Intel(R) Core(TM) i9-10980XE CPU and 32.0 GB of RAM. In contrast to the applications on the QT dataset, we did not use GPUs due to their RAM limitations, as we wanted to have a speed comparison as complete as possible.

Results of the speed evaluation are displayed in Figure 6.

Note that execution times could not be computed using the regular method for $N > 50000$ sampling points, due to the lack of RAM on the computer used. In contrast, the fast version of Trend Filtering did not store anything in memory besides the signal itself, and therefore could be used for all the cases. For the numerical estimation of the complexity, we show in this graph

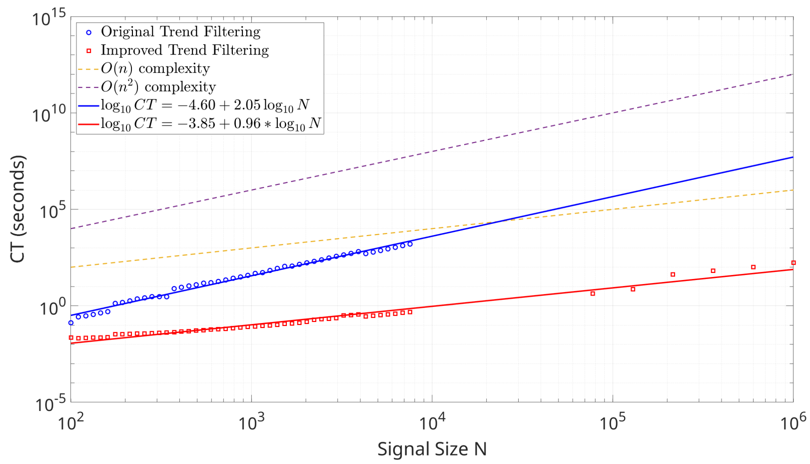


Figure 6: Average computation times (CT, in seconds) of regular Trend Filtering and its accelerated version as a function of the size of $\mathbf{y}$.

both the N and N^2 lines, and perform a linear regression (in the log-log scale) using the execution times obtained. The slope and intercept of both regression lines appear also in the figure's legend.

Based on the experiments, it can be clearly observed that the proposed acceleration of Trend Filtering outperforms by far the standard implementation. As shown both theoretically and numerically, the complexity went down from polynomial to linear, which results in a significant gain of speed. Moreover, the fact that we do not need to store any large matrix in memory in the fast implementation allows us to process larger signals, even in cases where the regular implementation fails. Consequently, the simulations show that FDP-TF can be made suitable for the processing of large signals in devices with memory limitations, thus opening the way for a future real-time implementation in wearable devices.

4.2. Evaluation on the QT Dataset

Since we can store at each iteration the history of the obtained zero-crossings, we investigated the following possible strategies to locate the peaks of the P-QRS-T waves:

- **Strategy 1:** Compute zero-crossings of $\hat{\boldsymbol{\theta}}(\lambda)$ only after the final iteration.

- **Strategy 2:** The zero-crossings of $\hat{\boldsymbol{\theta}}(\lambda)$ are stored in memory as iterations of the proximal gradient are performed, and are averaged. More precisely, in this case any given point k receives a score equal to the average number of sign changes of $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\lambda_2)$ (see Section 3.2) which have occurred so far at k. It is expected that points materializing waves endpoints typically score higher.

For both of the aforementioned strategies, the hyperparameters of the algorithms were set up as shown in Table 2.

Parameter	First pass	Second pass
λ	0.001	0.00001
Number of iterations	500	tested from 5000 to 100000
k	0	0
history step	N/A	5% of the total number of iterations

Table 2: Hyperparameters of FDP-TF for the first pass (baseline removal) and the second pass (peak localization).

In order to quantitatively assess the performances of the proposed approach, we follow the methodology described in [21] and consider that a difference greater than 120 ms (30 sampling points for the QT dataset) between the annotation and the output provided by our method is a missed detection. This allows us to compare the obtained results with other existing methods described in [21]. Results are always displayed for the first channel of the ECG recordings.

On the one hand, Figure 7 provides a qualitative example of the performance of FDP-TF on part of a typical ECG recording from the QT dataset, where the baseline wander can be clearly appreciated. The precise localization of the peaks of the P, R and T waves can be observed in all cases. On the other hand, the quantitative performance of FDP-TF can be appreciated in Figure 8 and Figure 9, which represent the precision of the peak localization as a function of the number of iterations for strategy 1 and strategy 2, respectively.

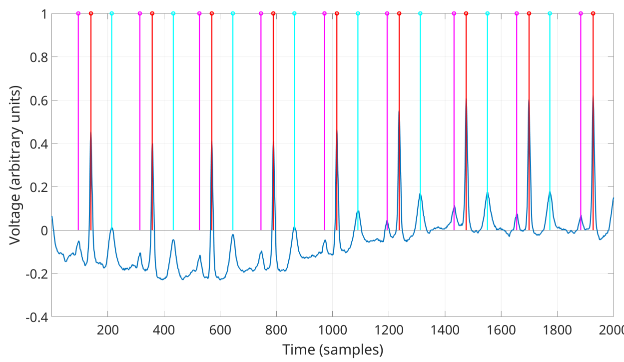


Figure 7: Result obtained with 30000 iterations on part of a QT dataset signal (strategy 1). QRS locations are displayed in red, P extrema in magenta and T extrema in cyan.

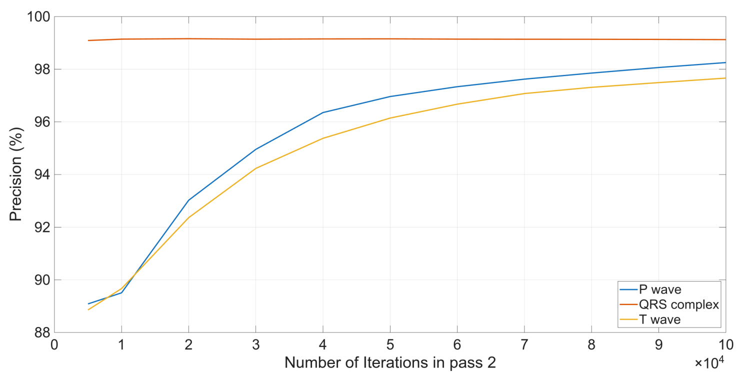


Figure 8: Precision of peak extrema localization for strategy 1. Precision values comparable to state-of-the-art methods are obtained from 40000 iterations.

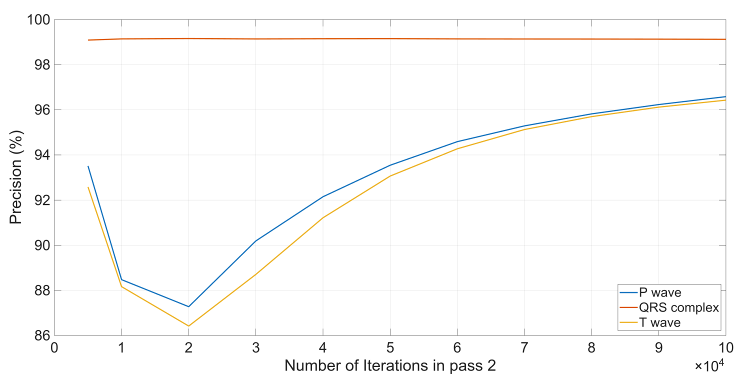


Figure 9: Precision of peak extrema localization for strategy 2.

For the sake of comparison, the results attained by FDP-TF on the QT dataset are summarized in Table 3 for both 60000 and 100000 iterations (strategy 1) and 100000 iterations (strategy 2).

Method	P-wave peak localization	QRS localization	T-wave peak localization
FDP-TF			
Strategy 1 (60000)	0.974	0.994	0.968
Strategy 1 (100000)	0.983	0.996	0.977
Strategy 2 (100000)	0.969	0.996	0.974
Laguna [6]	0.973	0.999	0.956
Martinez [13]	0.995	0.997	0.975
Singh [49]	0.960	N/A	0.91
Di Marco [14]	0.979	1	0.973
Sun [16]	0.998	1	0.967
Martinez [9]	0.978	1	0.978
Vazquez [10]	0.998	1	0.968
Vitek [11]	0.993	1	0.92
Hughes [15]	0.960	0.995	0.953
Han [27]	N/A	0.999	0.986
Peimankar [25]	N/A	0.997	0.977
Liu [28]	0.998	1	0.996
Li [34]	0.991	0.985	0.985

Table 3: Peak localization results attained by some state-of-the-art methods (results taken from Table 4 in [21] and Table 1 in [27]). Results for FDP-TF are presented for strategy 1 with 60000 and 100000 iterations and strategy 2 with 100000 iterations.

In this table, we also summarize the results obtained by other state-of-the-art methods described in [21], as well as several deep learning based methods. Comparing the performance of FDP-TF with other state-of-the-art methods, it can be seen that FDP-TF achieves a competitive performance for the location of all peaks. Although there are discrepancies in the computation of the performance metrics, available results from recent deep learning based methods [31, 32, 27, 28, 25, 34] confirm that FDP-TF attains competitive results: up to 98.3% in P peak detection for FDP-TF vs. $97.9 - 99.14\%$ for deep learning approaches; 99.6% in R peak (i.e., QRS complex) detection for FDP-TF vs. $98.48 - 99.9\%$ for deep learning approaches; and 97.7% for the T wave vs. $98.46 - 99.6\%$ for deep learning approaches. Moreover, let us remark that this good performance was achieved without the costly training process and the large computational storage required by deep learning methods.

Compared to classical methods which do not rely on deep learning, FDP-TF offers a more balanced and consistently high localization performance across all ECG peaks, as derived from Figure 8. While [14, 16, 9, 10] achieve perfect QRS localization (1.000) and strong results for P and T waves, FDP-TF trained using Strategy 1 with 100,000 iterations provides comparable QRS accuracy (0.996), while surpassing or matching T-wave localization in most cases. It also achieves higher P-wave accuracy than [14] and [15], which is crucial given the P wave's lower amplitude and more complex variability.

4.3. Discussion

4.3.1. Results analysis

With respect to the results, it can be noted that both strategies achieve excellent precision in locating the R wave ($> 99\%$), but strategy 1 can achieve better precision in locating the P and T waves than strategy 2, provided that the appropriate number of iterations is used. As shown in Figure 4, by stopping the algorithm too early (i.e., by using too few iterations) we miss relevant extrema. However, stopping it too late (i.e., by using too many iterations) is also an issue, since it leads to overfitting and thus some of the selected zero-crossings are related to noise and not to the actual peaks. A visual illustration of this fact is given is Figure 10.

It can be seen that when the number of iterations increases, the obtained regressor also focuses on the sign changes brought by the noise, which leads to an increased number of false detections. We found out empirically that a satisfactory peak localization with a reasonable false alarm rate could be attained for an intermediate number of iterations (30000-70000), with a precision comparable to state-of-the-art approaches achieved for 60000 iterations. For this number of iterations, a 15-minute long signal was processed in about 3 minutes using a GeForce RTX 2080Ti GPU. This opens the way for a practical, real-time implementation on GPU servers connected to wearable devices, for example.

On the other hand, when relying on the history (as done by strategy 2), Figure 9 shows that it is important to perform a large number of iterations to obtain results comparable to the state of the art. The slow increase of the curve seems to indicate that an even larger number of iterations could help attain better results than for strategy 1. However, this comes at the price of a longer execution time, which makes the proposed approach less attractive for real-time applications. The computation of the average of the iterations, nevertheless, remains a topic to be thoroughly investigated

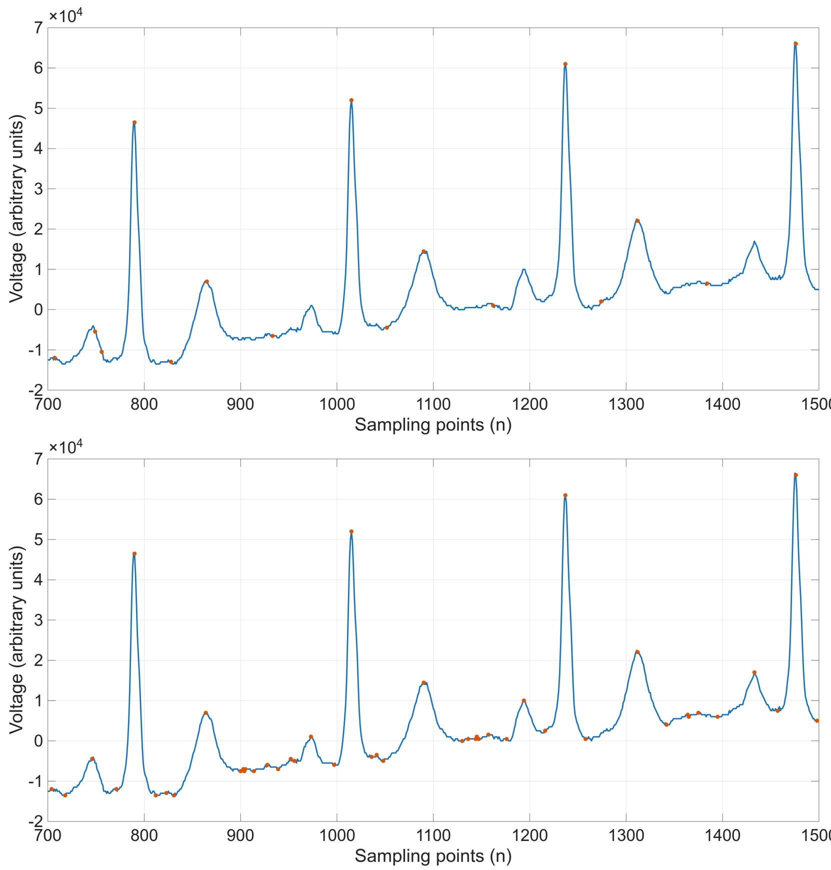


Figure 10: Part of one QT dataset recording and extracted candidate points for 5000 iterations (upper figure) and 100000 iterations (lower figure) of FDP-TF, using strategy 1.

further on. Indeed, not all the iterations have the same weight, as can be easily seen from Figure 4: some extrema are detected at very early stages of the iterations, but P waves are usually detected at a much later stage, similarly to the previous algorithm described in [36]. Moreover, at late stages, the estimation of the derivative provided by the second trend filtering pass becomes less sparse, and tends to estimate the changes brought by the noise as well. This means that even with a precision increase, the number of false positive detections can increase as well. Therefore, we infer that strategy 2 could be improved by performing a weighted average, giving more weights to the most relevant iterations of FDP-TF, which would allow us to locate peaks efficiently, but also onsets and offsets where the derivatives exhibit significant changes. These last points are out of the scope of the present contribution, which focuses on peak locations and algorithmic acceleration, and should be investigated in further contributions.

Another advantage of the proposed method is that FDP-TF maintains a balanced performance across all peaks, while most of the non-deep learning based methods usually favor QRS localization and provide lower performance at the T wave (in that matter, our method is second to best with [9]). Note also that the proposed approach depends on less hyperparameters than the ones relying on multiscale analysis [14, 16, 15] or digital signal processing approaches [10, 9]: indeed, these methods need numerous hand-tailored parameters such as depth and mother wavelet or basic structuring elements, decision rules to infer the peaks limits and extrema, etc. Thus FDP-TF provides a scalable and interpretable framework, achieving localization accuracy on par with state-of-the-art methods, with lower computational burden and offering more flexibility than the other approaches.

Finally, regarding the hyperparameters, we note that the choice of λ_2 (especially) might be challenging. Indeed, some P waves might go undetected for too high values of λ_2; on the other hand, smaller values of λ_2 yield an estimator which might be more sensitive to noise and small perturbations. Besides the previous remark, note also that there is a strong interconnection between λ (the sparsity parameter used) and the required number of iterations. Figure 4 seems to show that the local extrema of the waves are spotted very early. This seems to be corroborated by Figure 9, where averaging the history allows us to rely on much fewer iterations.

4.3.2. Interference influence

Since FDP-TF shares similarities with the original ATF proposed in [36], we expect it to share as well similar properties with respect to baseline wander, electrode noise motion, and EMG noise. These aspects were thoroughly investigated in [36] for ATF. Though a full analysis on the QT dataset is not in the scope of the present manuscript (these recordings have mostly an excellent SNR), we can illustrate these aspects graphically on some examples. First of all, Figure 11 illustrates the results obtained with a segment including a significant baseline wander, initially present in the recording.

The figure, when compared to the ones in [36], illustrates that the modifications introduced in FDP-TF do not compromise its performance in presence of baseline wander noise. Then, ECG noises from the MIT-BIH Noise Stress Test Database [50], available on the Physionet website, were added to

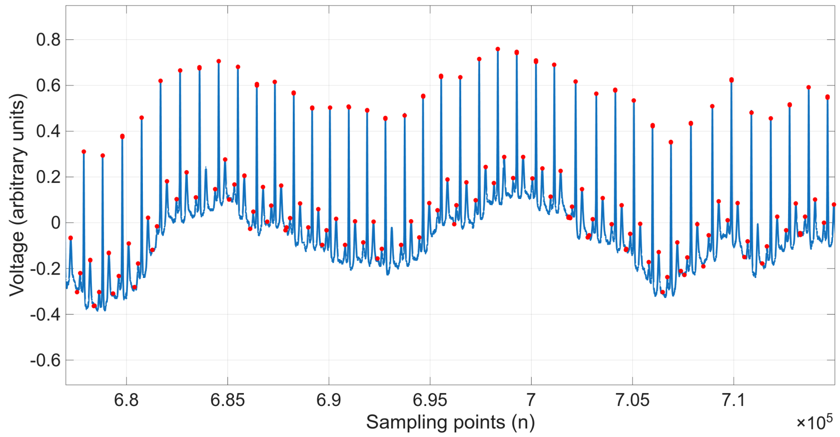


Figure 11: Influence of the baseline wander on the results

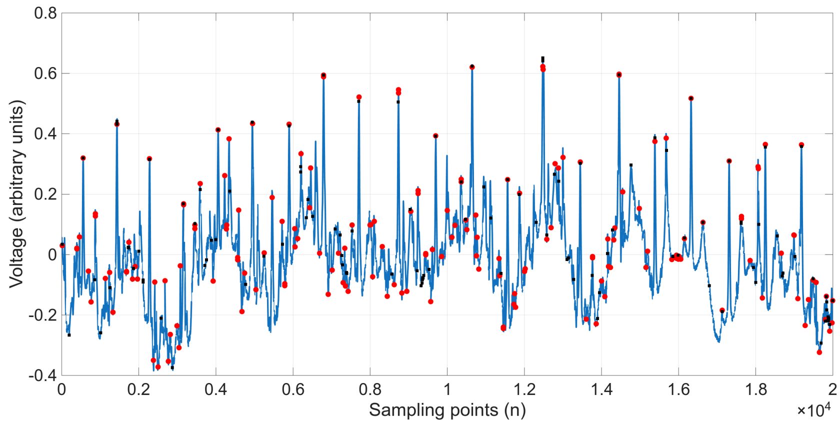


Figure 12: Influence of the electrode motion noise on the fiducial points candidates. Black squares: original extrema candidates on the noise-free signal. Red dots: results after addition of electrode motion noise.

one original recording to test other noise types. Figures 12 and 13 show the fiducial points candidates extracted from one recording with 5000 iterations of FDP-TF after addition of electrode noise and EMG noise, respectively. Indeed, the examples displayed in Figures 12 and 13 seem to illustrate that, similarly to ATF, the additional noise adds additional peak candidates, which makes the peak localization more difficult. For electrode noise, the number of additional candidate points is 695, whereas for EMG noise, it increases up to 1797. This last count shows that, similarly to ATF, EMG noise can be the most problematic noise to tackle with FDP-TF, as the false alarm rate can

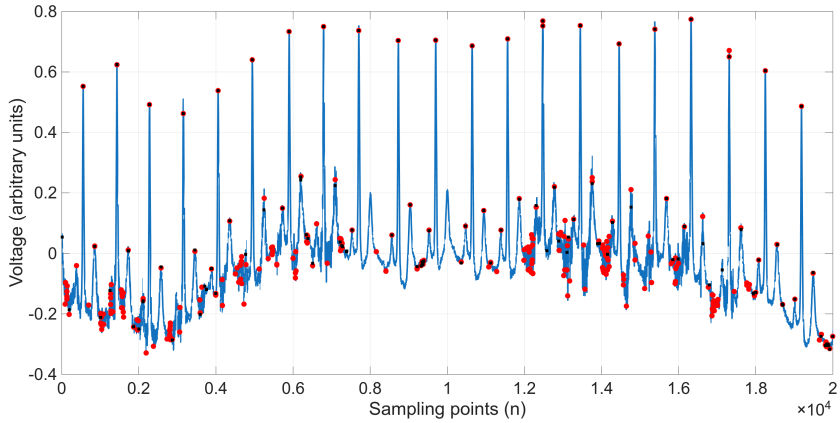


Figure 13: Influence of the EMG noise on the fiducial points candidates. Black squares: original extrema candidates on the noise-free signal. Red dots: results after addition of electrode motion noise.

significantly increase. However, in both cases, candidate peaks are selected close to the ones found for the noise-free signal. This seems to indicate that for reasonable SNRs, the proposed approach could still provide satisfactory results. This last point, as well as the use of preliminary denoising techniques to attenuate the noise influence, will be investigated in future contributions.

4.3.3. Influence of the sampling frequency

As more advanced ECG devices provide recordings sampled at higher frequencies, it is also important to investigate the role of the sampling frequency in the results. To illustrate this point, we present in Figure 14 results on a part of a QT dataset recording (sampled at 250 Hz), and the results obtained on the same segment which underwent an interpolation of factor 4, to simulate a sampling performed at 1kHz, all the other hyperparameters and number of iterations (10000) being equal in both cases. For both results, we display the located points using averaging of the zero-crossings of $\hat{\boldsymbol{\theta}}(\lambda_2)$ over the full iterations' history.

As aforementioned, since averaging is done on the full history of iterations, small peaks will be likely undetected, which results on poor peak localization. However, we note that the peaks are better located with a higher sampling frequency. We infer that the higher the sampling frequency, the better the results of the estimation of the derivative using Trend Filtering are, therefore yielding better peak localization. Though a more thorough investigation is required for validation, this result seems to show that the method could also

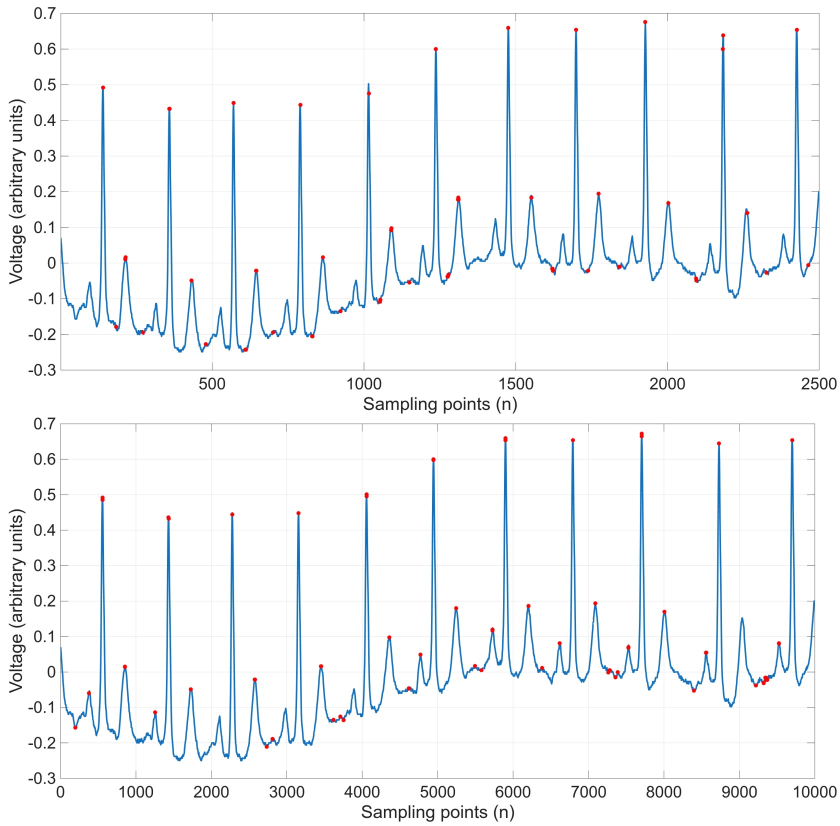


Figure 14: Original (up) and oversampled (down) signal, and results obtained in peak localization (red dots) with 10000 iterations. An increase in the frequency sampling tends to improve performance.

be successfully applied for more recent ECG systems, and could yield better results with fewer iterations.

5. Conclusions and Future Lines

This paper has presented Fast Double-Pass Trend Filtering , a complete reformulation of the ATF algorithm originally developed by the authors that has resulted in a significantly faster method for the localization of peaks in ECG recordings. The results presented have demonstrated that the method is one order of magnitude faster and, since no matrix storage in memory is required, it can be used for recordings of arbitrary length and even real-time processing using an entry-level GPU. The results in the QT dataset have shown similar performance to other well-established methods, especially in the case of the T-wave peak localization. The focus of this paper has been the design of a fast algorithm, capable of real-time peak detection in long ECG recordings, with a competitive performance with respect to more costly DL-based approaches. A post-processing approach (e.g., based on zero-crossings) could then be used to approximately detect the waves onset and offset. Its integration within the proposed framework will be detailed in future works, in order to have a complete and robust full wave segmentation framework. The evaluation of FDP-TF on other standard datasets and a more comprehensive investigation of the effect of the algorithm's hyperparameters (especially the sparsity parameter λ) will also be considered.

Acknowledgment

This work has been funded by MICIU/AEI/10.13039/501100011033 and by ERDF "A way of making Europe" under the grant agreement No. PID2023-153035NB-I00.

References

[1] L. Sornmo, P. Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications, Elsevier, 2005.

[2] G. D. Clifford, F. Azuaje, P. E. McSharry, Advanced Methods and Tools for ECG Data Analysis, Artech House, 2006.

[3] A. Gacek, W. Pedrycz, ECG signal processing, classification and interpretation: a comprehensive framework of computational intelligence, Springer Science & Business Media, 2011.

[4] J. P. do Vale Madeiro, P. C. Cortez, J. M. D. S. Monteiro Filho, A. R. A. Brayner, Developments and applications for ECG signal processing: Modeling, segmentation, and pattern recognition, Academic Press, 2018.

[5] D. Bentaleb, Z. Khatar, Multi-criteria bayesian optimization of empirical mode decomposition and hybrid filters fusion for enhanced ecg signal denoising and classification: Cardiac arrhythmia and myocardial infarction cases, Computers in Biology and Medicine 184 (2025) 109462. doi:https://doi.org/10.1016/j.compbiomed.2024.109462.

[6] P. Laguna, R. Jane, P. Caminal, Automatic detection of wave boundaries in multilead ECG signals: validation with the CSE database, Computers and Biomed. Research 27 (1994) 45–60.

[7] I. Christov, A review of electrocardiogram filtering, Journal of Electrocardiology 43 (2010) 486–496.

[8] V. Krasteva, I. Jekova, T. Stoyanov, R. Schmid, In search of an optimal FIR filter for ECG delineation, in: 2024 Computing in Cardiology (CinC), 2024.

[9] A. Martinez, R. Alcaraz, J. Rieta, Application of the phasor transform for automatic delineation of single-lead ECG fiducial points, Physiological Measurement 31 (2010) 1467–1485.

[10] C. R. Vazquez-Seisdedos, J. E. Neto, E. J. Maranon Reyes, A. Klautau, R. C. Limao de Oliveira, New approach for T-wave end detection on electrocardiogram: Performance in noisy conditions, BioMedical Engineering OnLine 10 (2011) 77.

[11] M. Vítek, J. Hrubeš, J. Kozumplík, A wavelet-based ECG delineation in multilead ECG signals: Evaluation on the CSE database, in: World Congress on Medical Physics and Biomedical Engineering, 2009, pp. 177–180.

[12] N. Spicher, M. Kukuk, Delineation of electrocardiograms using multiscale parameter estimation, IEEE Journal of Biomedical and Health Informatics 24 (2020) 2216–2229.

[13] J. Martinez, R. Almeida, S. Olmos, A. Rocha, P. Laguna, A wavelet-based ECG delineator: evaluation on standard databases, IEEE Transactions on Biomedical Engineering 51 (2004) 570–581.

[14] L. Y. Di Marco, L. Chiari, A wavelet-based ECG delineation algorithm for 32-bit integer online processing, Biomedical Engineering Online 10 (2011) 23.

[15] N. Hughes, L. Tarassenko, S. J. Roberts, Markov Models for Automated ECG Interval Analysis, in: Advances in Neural Information Processing Systems, volume 16, MIT Press, 2003.

[16] Y. Sun, K. L. Chan, S. M. Krishnan, Characteristic wave detection in ECG signal using morphological transform, BMC Cardiovascular Disorders 5 (2005) 28.

[17] S. Monzón, T. Trigano, D. Luengo, A. Artés-Rodríguez, Sparse spectral analysis of atrial fibrillation electrograms, in: IEEE International Workshop on Machine Learning for Signal Processing (MLSP), 2012, pp. 1–6.

[18] T. Trigano, D. Luengo, Intracardiac ECG pulse localization using overlapping block sparse reconstruction, Biomedical Signal Processing and Control 79 (2023) 103921.

[19] D. Luengo, J. Vía, S. Monzón, T. Trigano, A. Artés-Rodríguez, Cross-Products LASSO, in: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2013, pp. 6118–6122.

[20] D. Luengo, J. Vía, T. Trigano, Efficient iteratively reweighted LASSO algorithm for cross-products penalized sparse solutions, in: 28th European Signal Processing Conference (EUSIPCO), IEEE, 2021, pp. 2045–2049.

[21] I. Beraza, I. Romero, Comparative Study of Algorithms for ECG Segmentation, Biomedical Signal Processing and Control 34 (2017) 166–173.

[22] R. Huo, L. Zhang, F. Liu, Y. Wang, Y. Liang, S. Wei, Ecg segmentation algorithm based on bidirectional hidden semi-markov model, Computers in Biology and Medicine 150 (2022) 106081. doi:https://doi.org/10.1016/j.compbiomed.2022.106081.

[23] J. Wang, R. Li, R. Li, B. Fu, A knowledge-based deep learning method for ECG signal delineation, Future Generation Computer Systems 109 (2020) 56–66.

[24] I. Sereda, S. Alekseev, A. Koneva, R. Kataev, G. Osipov, ECG segmentation by neural networks: Errors and correction, in: 2019 International Joint Conference on Neural Networks (IJCNN), IEEE, 2019, pp. 1–7.

[25] A. Peimankar, S. Puthusserypady, DENS-ECG: a deep learning approach for ECG signal delineation, Expert Systems with Applications 165 (2021) 113911.

[26] A. N. Londhe, M. Atulkar, Semantic segmentation of ECG waves using hybrid channel-mix convolutional and bidirectional LSTM, Biomedical Signal Processing and Control 63 (2021) 1–11.

[27] C. Han, W. Que, S. Wang, J. Zhang, J. Zhao, L. Shi, QRS complexes and T waves localization in multi-lead ECG signals based on deep learning and electrophysiology knowledge, Expert Systems with Applications 199 (2022) 117187.

[28] J. Liu, Y. Jin, Y. Liu, Z. Li, C. Qin, X. Chen, L. Zhao, C. Liu, A novel p-qrs-t wave localization method in ecg signals based on hybrid neural networks, Computers in Biology and Medicine 150 (2022) 106110. doi:https://doi.org/10.1016/j.compbiomed.2022.106110.

[29] X. Liang, L. Li, Y. Liu, D. Chen, X. Wang, S. Hu, J. Wang, H. Zhang, C. Sun, C. Liu, Ecg_segnet: An ecg delineation model based on the encoder-decoder structure, Computers in Biology and Medicine 145 (2022) 105445. doi:https://doi.org/10.1016/j.compbiomed.2022.105445.

[30] A. Atienza, J. Bardram, S. Puthusserypady, An efficient and flexible deep learning method for signal delineation via keypoints estimation, arXiv preprint arXiv:2407.20258 (2024).

[31] V. Krasteva, T. Stoyanov, R. Schmid, I. Jekova, Delineation of 12-lead ecg representative beats using convolutional encoder–decoders with residual and recurrent connections, Sensors 24 (2024) 4645.

[32] S. Nurmaini, A. Darmawahyuni, M. N. Rachmatullah, J. Effendi, A. I. Sapitri, F. Firdaus, B. Tutuko, Beat-to-beat electrocardiogram waveform classification based on a stacked convolutional and bidirectional long short-term memory, IEEE Access 9 (2021) 92600–92613. doi:10.1109/ACCESS.2021.3092631.

[33] A. Darmawahyuni, S. Nurmaini, M. N. Rachmatullah, P. P. Avi, P. Teguh, S. A. I., B. Tutuko, F. Firdaus, Improved delineation model of a standard 12-lead electrocardiogram based on a deep learning algorithm, BMC Medical Informatics and Decision Making 23 (2023) 233.

[34] X. Li, W. Cai, B. Xu, Y. Jiang, M. Qi, M. Wang, SEResUTer: a deep learning approach for accurate ECG signal delineation and

atrial fibrillation detection, Physiological Measurement 44 (2023). doi:10.1088/1361-6579/ad02da.

[35] L. Chen, Z. Jiang, J. Barker, H. Zhou, F. Schlindwein, W. Nicolson, G. Ng, X. Li, Ecgvednet: A variational encoder-decoder network for ecg delineation in morphology variant ecgs, IEEE Transactions on Biomedical Engineering 71 (2024) 2143–2153.

[36] T. Trigano, S. Talala, D. Luengo, Adaptive trend filtering for ECG denoising and delineation, IEEE Journal of Biomedical and Health Informatics 27 (2023) 5755–5766. doi:10.1109/JBHI.2023.3314983.

[37] S.-J. Kim, K. Koh, S. Boyd, D. Gorinevsky, L1 Trend Filter, SIAM Review 51 (2009) 339–360.

[38] S. Boyd, N. Parikh, E. Chu, B. Peleato, J. Eckstein, Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers, Now Publishing, 2011.

[39] A. Ramdas, R. J. Tibshirani, Fast and Flexible ADMM Algorithms for Trend Filtering, Journal of Computational and Graphical Statistics 25 (2016) 839–858.

[40] N. Parikh, S. Boyd, Proximal Algorithms, Now Publishing, 2014. doi:10.1561/2400000003.

[41] A. Goldberger, L. Amaral, L. Glass, J. Hausdorff, P. Ivanov, R. Mark, J. Mietus, G. Moody, C. Peng, H. Stanley, PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals, Circulation 101 (2000) e215 – e220.

[42] G. Steidl, S. Didas, J. Neumann, Splines in Higher Order TV Regularization, International Journal of Computer Vision 70 (2006) 214–255.

[43] R. J. Tibshirani, Adaptive piecewise polynomial estimation via trend filtering, The Annals of Statistics 42 (2014) 285 – 323.

[44] R. Tibshirani, Regression Shrinkage and selection via the LASSO, Journal of the Royal Statistical Society 58 (1996) 267–288.

[45] A. Beck, M. Teboulle, A fast iterative Shrinkage-Thresholding algorithm for linear inverse problems, SIAM Journal on Imaging Sciences 2 (2009) 183.

[46] A. Beck, M. Teboulle, Gradient-based algorithms with applications to signal recovery, in: D. P. Palomar, Y. C. Eldar (Eds.), Convex optimization in signal processing and communications, Cambridge university press, Cambridge, UK, 2010, pp. 42–88.

[47] P. Laguna, R. Mark, A. Goldberger, G. Moody, A database for evaluation of algorithms for measurement of QT and other waveform intervals in the ECG, in: Computers in Cardiology (CinC), volume 24, 1997, pp. 673–676.

[48] P. McSharry, G. Clifford, L. Tarassenko, L. Smith, A dynamical model for generating synthetic electrocardiogram signals, IEEE Transactions on Biomedical Engineering 50 (2003) 289–294. doi:10.1109/TBME.2003.808805.

[49] Y. N. Singh, P. Gupta, ECG to Individual Identification, in: 2008 IEEE Second International Conference on Biometrics: Theory, Applications and Systems, 2008, pp. 1–8.

[50] G. B. Moody, W. Muldrow, R. G. Mark, A noise stress test for arrhythmia detectors, Computers in cardiology 11 (1984) 381–384.